

Classification Models Performance Report

Exploratory Data Analysis and Metrics Evaluation

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Abstract

This report presents an exploratory data analysis (EDA) and a comparative evaluation of four machine learning classification models—Logistic Regression, Random Forest, LightGBM (LGBM), and XGBoost—applied to sentiment analysis on the IMDB movie review dataset. The dataset consists of 50,000 balanced reviews labeled as positive or negative, with no missing values. Text data were transformed using two vectorization techniques: Bag of Words (BoW) and Term Frequency–Inverse Document Frequency (TF-IDF). Model performance was assessed using key evaluation metrics, including Accuracy, Precision, Recall, F1-Score, and AUC-ROC. The results demonstrate that Logistic Regression combined with TF-IDF achieved the best overall performance, presenting high accuracy and a near-optimal ROC curve. Additionally, bigram analysis provided insights into common expressions associated with positive and negative sentiments, highlighting linguistic patterns that influence classification. These findings reinforce the effectiveness of simpler linear models when combined with appropriate text representation techniques in sentiment classification tasks.

1 About the dataset

The IMDB dataset is a large dataset of film reviews. It is essentially a benchmark for classifying binary sentiments introduced in the article "Learning Word Vectors for Sentiment Analysis".

This dataset contains 50,000 polarized film reviews, extracted from IMDB. It is one of the leading datasets for NLP (Natural Language Processing) text classification.

DATASET <https://aclanthology.org/P11-1015/>.

2 Exploratory Data Analysis (EDA)

Exploratory Data Analysis (EDA) is a method of analyzing datasets to understand their main characteristics. The main objectives behind EDA include:

1. Data Quality: Identifying missing values, duplicates, and errors that require treatment.
2. Anomaly Detection: Finding outliers (discrepant values) that can distort future analyses and models.
3. Correlations and Relationships: Discovering how variables interact with each other, revealing dependencies or the lack thereof.

2.1 Data Distribution

EDA concluded that there are 50.000 records, distributed between 2 columns (review and sentiment). Neither has null values. The classes are balanced, 50% positive and 50% negative. The EDA explain the two columns has 50.000 registers no null. Thus, for each column there was an associated record.

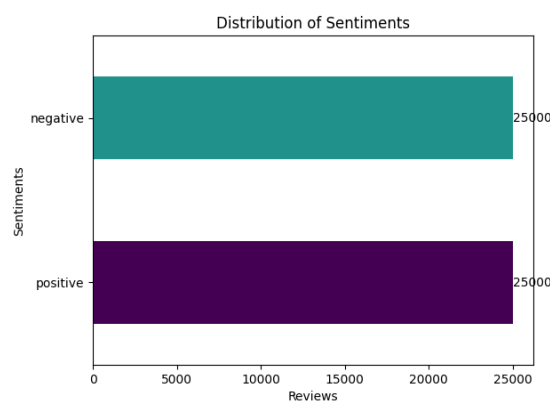


Figure 1: class balancing

3 Machine Learning

A machine learning model is a computational program trained on data to recognize patterns, make decisions, or generate predictions.

3.1 Machine Learning algorithms

Classification and Regression are supervised machine learning algorithms. The base algorithm for this project was Classification, which produces qualitative output where classes can be categorical, nominal, and ordinal.

3.1.1 Logistic Regression

Logistic Regression is a Machine Learning model mainly used for classification problems, especially when the target outcome is binary, such as “yes or no” or “purchase or no purchase.” It estimates the probability that an observation belongs to a certain class by using a function that converts results into values between 0 and 1. Its main advantage is that it is simple, fast, and easy to interpret, making it possible to understand the impact of each variable on the prediction. However, it may have limited performance when the data contains very complex or non-linear relationships.

3.1.2 Random Forest

Random Forest is a Machine Learning algorithm based on the combination of multiple decision trees, used for both classification and regression tasks. It works by creating several trees using random samples and random subsets of variables from the data, then combining their results to produce a more robust and accurate prediction. Its main advantage is that it handles complex and non-linear relationships well, while also reducing the risk of overfitting compared to a single decision tree. However, it can be less interpretable and more computationally demanding than simpler models.

3.1.3 LightGBM (LGBM)

LightGBM is a type of GBM (Gradient Boosting Machine) that is developed to provide faster training time and better performance than XGBoost. It uses a leaf-wise growth strategy instead of level-wise growth. Leaf-wise growth strategy involves growing the tree by pre-defining a maximum depth or a minimum number of observations for the leaf nodes of the tree. In this strategy, the tree depth and number of leaf nodes are pre-defined, and the tree is grown until it reaches the specified depth or number of leaf nodes. This strategy is more efficient than the

level-wise growth strategy, and it reduces the memory usage and creation time of the tree.

3.1.4 XGBoost

XGBoost is a popular option for many machine learning problems, especially for classification and regression of structured data. It also supports the use of multiple processors to speed up model training. Thanks to these features, XGBoost provides high accuracy, speed, and scalability.

4 Classification Metrics

4.1 Accuracy

Accuracy measures the proportion of correct predictions made by the model among all predictions. It is calculated by dividing the number of correct classifications by the total number of observations. Although it is a simple and intuitive metric, accuracy may not be the best option when the dataset is imbalanced, because a model can achieve high accuracy by mostly predicting the majority class.

4.2 Precision

Precision measures how many of the instances predicted as positive were actually positive. It is useful when the cost of false positives is high, meaning that it is important to avoid classifying negative cases as positive. A high precision indicates that the model makes fewer false positive predictions.

4.3 Recall

Recall measures how many of the actual positive instances were correctly identified by the model. It is especially important when the cost of false negatives is high, meaning that missing a positive case can be a serious problem. A high recall indicates that the model is good at finding most of the positive cases.

4.4 F1-Score

The F1-score is the harmonic mean between precision and recall. It is useful when there is a need to balance both metrics, especially in imbalanced datasets. A high F1-score means that the model has a good balance between correctly identifying positive cases and avoiding false positive predictions.

4.5 AUC-ROC

The area under the ROC curve (AUC) represents the probability that the model, given a randomly chosen positive and negative example, will rank the positive example better than the negative one.

The model with the larger area under the curve is generally the best.

5 Libraries and Imports

In this project, essential libraries were used for data analysis, visualization, text preprocessing, and Machine Learning model development. `pandas` was used to manipulate and organize the dataset, while `matplotlib` and `seaborn` helped create charts and visualizations. The `scikit-learn` library provided tools for splitting the data into training and testing sets, transforming text with `CountVectorizer` and `TfidfVectorizer`, encoding labels, training models such as Logistic Regression and Random Forest, and evaluating performance using metrics such as accuracy, confusion matrix, classification report, and AUC-ROC. The models `XGBoost` and `LightGBM` were also used, as they are efficient boosting algorithms for classification tasks. For text processing, the `NLTK` library was applied with modules for stopwords removal, tokenization, and stemming, together with `re` for regular expressions. Finally, the `time` package was used to measure the training time of each model.

The models were trained and tested using standard splitting protocols. Performance was measured using Accuracy, Precision, Recall, and F1-Score.

6 Results

In this training, I used two vectorization tools: TF-IDF and BoW.

TF-IDF: This is a statistical measure used in Natural Language Processing and Data Mining to evaluate the importance of a word in a document relative to a collection (corpus). It is fundamental for converting text into numbers.

BoW (Bag of Words): This is a simple Natural Language Processing (NLP) and Machine Learning technique used to convert text into numbers. It creates a "vocabulary" with all the unique words in a text and counts the frequency of each one, ignoring grammar and order.

The table below shows the results of the metrics: Accuracy, AUC-ROC, Recall, and F1-Score for each classification algorithm, in addition to comparing these algorithms with different vectorization tools used (TF-IDF and BoW).

Experiment	Accuracy	AUC-ROC	Precision	Recall	F1-Score	Training time (s)	Prediction time (s)
Random Forest (TF-IDF)	84.98%	0.9269	84.99%	84.98%	84.98%	114.82	1.21
LightGBM (TF-IDF)	87.11%	0.9462	87.12%	87.11%	87.11%	99.61	0.31
XGBoost (TF-IDF)	85.50%	0.9331	85.53%	85.50%	85.50%	103.43	0.08
Logistic Regression (TF-IDF)	88.70%	0.9545	88.72%	88.70%	88.70%	2.61	0.01
Random Forest (BoW)	84.71%	0.9228	84.72%	84.71%	84.71%	103.36	0.66
LightGBM (BoW)	87.23%	0.9475	87.24%	87.23%	87.23%	20.65	0.30
XGBoost (BoW)	85.37%	0.9344	85.41%	85.37%	85.37%	6.89	0.22
Logistic Regression (BoW)	86.83%	0.9353	86.83%	86.83%	86.83%	5.24	0.01

Figure 2: metrics and time

Based on these metrics, it is possible to identify which classification algorithms perform best. In this case, the Logistic Regression model, using TF-IDF as a vector, presented the best results compared to the others. The ROC curve below shows that this same model has a result close to 1, having the largest area under the curve.

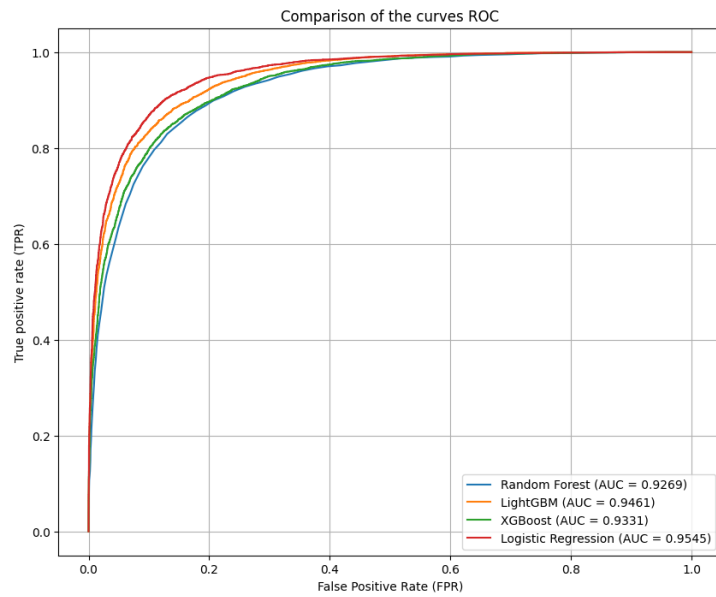


Figure 3: AUC-ROC

Furthermore, another analysis performed relates to the bigrams most commonly used by users when conducting their reviews. But what exactly are bigrams?

A bigram is a continuous sequence of two adjacent words (tokens) found in a text.

In this case, the positive bigrams that were most frequently presented in the analysis are shown in the table below, along with the frequency with which they appeared:

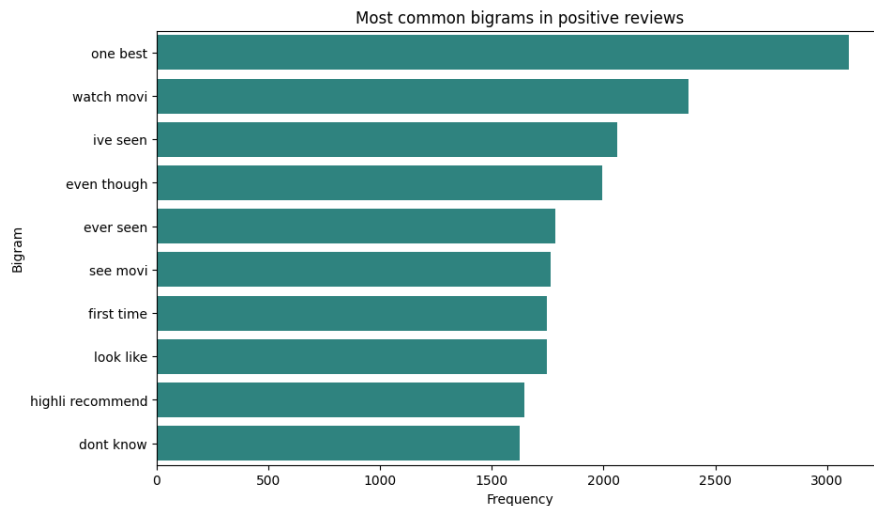


Figure 4: Positives bigrams most common

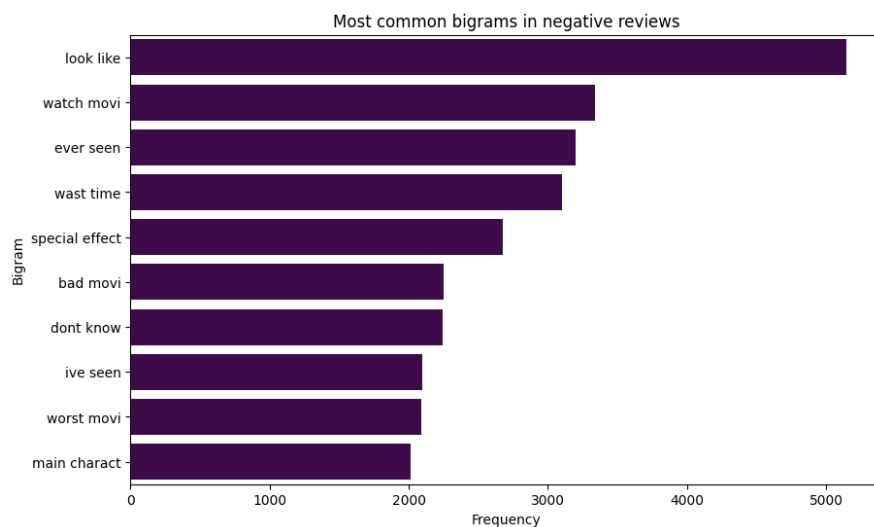


Figure 5: Negatives bigrams most common

Analysis of the most frequent bigrams reveals clear differences between positive

and negative reviews. In negative comments, directly critical expressions predominate, such as "bad movie," "worst movie," and "waste time," in addition to terms related to technical aspects, indicating that users tend to emphasize flaws and problematic points. On the other hand, in positive reviews, expressions such as "one best" and "highly recommend" appear, reflecting satisfaction and explicit recommendation, as well as terms associated with memorable experiences.

It is also possible to observe the presence of some bigrams in both classes, such as "watch movie" and "ever seen." These terms are considered neutral, as they do not carry sentiment on their own and depend on the context in which they appear. This shows that not all bigrams are useful for distinguishing sentiment.

7 Conclusion

In conclusion, sentiment analysis using machine learning models proves to be an effective approach for extracting meaningful insights from textual data, particularly in business and user feedback contexts. The comparative analysis shows that, despite the availability of more complex models such as Random Forest, LightGBM, and XGBoost, Logistic Regression achieved superior performance when combined with TF-IDF vectorization. This indicates that simpler models can perform exceptionally well when the data is properly preprocessed and represented. Furthermore, the analysis of bigrams revealed clear linguistic differences between positive and negative reviews, demonstrating the importance of contextual features in sentiment classification. Overall, this study highlights the relevance of model selection, feature engineering, and evaluation metrics in building efficient and reliable sentiment analysis systems. Future work may explore deep learning approaches, hyperparameter tuning, and the use of more advanced text representations to further improve performance.

8 References

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